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Prototype

VR Simulator

Ansat helicopters



Preface

This free software product has a minimal configuration and is intended solely for testing purposes. It can be used in two modes:

1. Display
2. VR.

If the program detects a connected VR headset upon launch, it will launch in VR mode. Otherwise, it will automatically launch in display mode. Display mode provides the following interfaces:

1. Mouse.
2. Keyboard
3. Joystick - T.Flight Stick X.
You can interact with the program either through the menu or directly - using the mouse and keyboard.
In VR mode, everything is controlled with controllers.

Installation

The installation process is described in the ReadMe.txt file of the installation package you downloaded.

To uninstall the software package, use unins000.exe, which is located in the folder where you installed the simulator.

Minimum configuration for VR:
Core i7 processor and 16 GB of RAM;
RTX 2070 graphics card (or equivalent performance).
At least 5 GB of free disk space.
A less demanding configuration may suffice for non-VR mode.

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Launch

The simulator is launched by double-clicking on the shortcut on the desktop, which will appear there after you install the program.

Upon launch, two windows appear. One displays information about running applications, and the other displays the main simulator window. It is strongly recommended that you do not interact with this window until the program has fully loaded.

Once loaded, the window can only be resized using the keyboard. ALT+Enter will enlarge or reduce the window. The rest of the interface, except for closing the window, is locked.

You can close the simulator either through the menu item or by closing the information window that was launched first.

I recommend that you first familiarize yourself with the program in display mode. The functionality in VR and display mode is absolutely identical. VR mode is more spectacular and gives the illusion of reality.

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INTERFACES IN MODE "MONITOR"

To start controlling the simulator, place the cursor on the window and press the left or right mouse button.

moving forward	**-W key**
moving backwards	**-S key**
move right	**-D key**
move left	**-A key**
Turn	**executed by the mouse when it is active in the window**
Move inside cabins	**key - I**
Move from the cabin to environment	**-O key**

To detach the cursor from the simulator window, press the X key and, while holding it down, press the left or right mouse button, and then you can move it to another window.

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Movements inside the cabin

except for the WSDA keys Q key - above E key - below

Move from the cabin to environment	**key - O**
You can also move discretely to the right, left, forward, backward	**arrow ---> arrow <--- up arrow up arrow**
To the lower panels	**key - U**
To the top panels	**key - Y**
Move to the original view inside the cabin	**key - I**

The same can be done
via the menu, the entrance to which you will see in the upper left corner of the window.

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Interacting with switches in the cockpit

If the cursor is in the window and you press the X key, then when you hover the mouse over a control element, the cursor changes its appearance from an arrow to a hand

when the hand appeared:

If the switch is two-position, then when you press the left or right mouse button, the switch will change its position to the opposite.

If the switch has three or more positions, then when you press

- left mouse button, it will move to the left or down

- right mouse button, it will move to the right or up

If the switch has three or more positions, then when you press

- left mouse button, it will move to the left or down

- right mouse button, it will move to the right or up

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Interacting with helicopter controls using mouse and keyboard.

click the central wheel (with the "X" released)
mice include control of the ROSH and RTSSH mouse and pedals.

Clicking the central mouse wheel again
(while releasing the "X") turns off the mouse
control of the ROSH and RTSSH mouse and pedals.

If you click the central mouse wheel again (while holding down "X"), it also turns off the ROSH and RTSSH control and

pedals, but the RTS is installed in neutral position

To control the pedals

"X" + "C" left pedal forward "X" + "V" right pedal forward

If the "X" key is not pressed, you can control the pedals,

by clicking the left or right mouse button.

To manage ROSH

mouse wheel towards you - up
mouse wheel away from you - down

To control the RCS

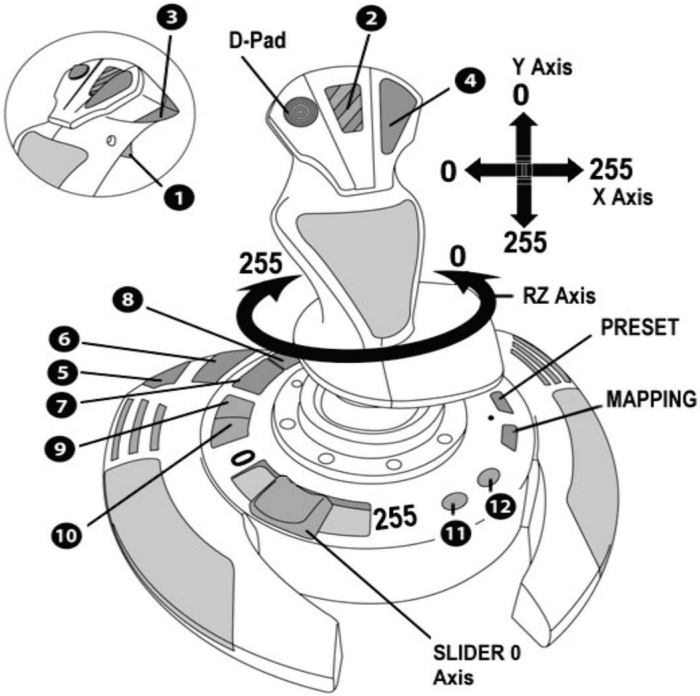
"X" + moving the mouse on the pad changes position of the RCS

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The interactions with will be described below.

helicopter simulator controls using a JOYSTICK

T.FLIGHT STICK X in the simulator operating mode NOT in VR



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Help with the ANSAT helicopter simulator, with a hydromechanical control system

Before starting the simulator, set the "PC-PS" switch on the joystick to "PC" and connect it to your computer. Set the joystick axes to neutral. The indicator on the joystick between the Preset and Mapping buttons should be red.

To start controlling the joystick, you need to press the "J" key

If a red message appears indicating that the joystick is in the wrong configuration, move the SLIDER AXIS (which controls the ROSH) to one of the extreme positions until you see a green message "Joystick configured."

The X and Y axes control the throttle, and the RZ axis controls the pedals. Button number 1 is the throttle trimmer.

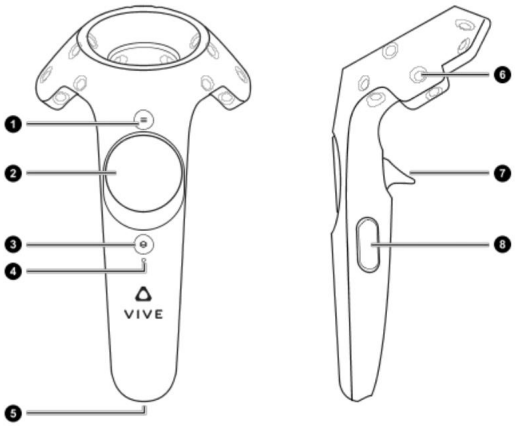
To disable joystick control, press the "J" key.

In VR mode, joystick control is enabled: Main Menu (on the right hand) -> Menu Item -> Enable Joystick Control. Disabling is achieved by pressing the same menu item again.

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VR INTERFACES

Moving in the environment - outside the cabin.



1	Кнопка «Меню»
2	Сенсорная панель
3	Кнопка «Система»
4	Индикатор состояния
5	Порт micro-USB
6	Датчик отслеживания
7	Курок
8	Кнопка «Захват»

Left Controller - Pressing the left half of the touchpad initiates a left turn.

Left controller - pressing the right half of the touchpad initiates a right turn.

Right controller - touching the half of the touchpad away from you initiates teleportation forward.

Right controller - press the one farthest from you half of the touchpad initiates a smooth forward movement.

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Moving inside the helicopter cabin.

Left Controller - Pressing the left half of the touchpad initiates a left turn inside the cabin.

Left Controller - Pressing the right half of the touchpad initiates a right turn inside the cabin.

The right controller - its touchpads - are locked.

Movement inside the cockpit (adjustment of landing site)
Main menu (on the right hand) -> menu item -> adjustment pilot sitting

Moving from the cockpit to the outside and back to the cockpit
Main menu (on the right hand) -> menu item -> Go to pilot cabin

turn on control of the helicopter's levers - joystick
Main menu (on the right hand) -> menu item - Joystick is off

Turn on helicopter pedal control (main menu
(on the right hand) -> Pedal control off)

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Interacting with switches in cabin

The switches are operated using the index fingertip of both hands. On the left hand, the contact area is highlighted by a paralepiped.

Two-position switches are activated by simply touching them. Three-position switches are also activated from their extreme positions by touching them, but from the neutral position, they are moved up (to the right) or down (to the left), using the same motion we would do in real life. For example,
to move a three-position switch from the neutral position down, you would slide it from top to bottom, just as we would do in real life.

Interaction with multi-position switches is achieved by simply touching the tip

index finger. A left-hand touch moves the switch to the left, while a right-hand touch moves it to the right. These directions can be reversed by pressing and releasing the grip button on either controller when they're not being gripped or touched with the index finger, etc.

Rotary control knobs, similar to radio volume controls (and anything else that needs to rotate). Simply touch and hold them (they'll light up), and they'll rotate until you touch them or they reach their full stop.

You can change the direction of rotation in the manner described in the previous paragraph.

There are several other ways to make control more comfortable, but they will be described later.

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Interaction with helicopter controls.

You can interact with the helicopter's controls in VR mode using a joystick, or, if a joystick isn't connected, using VR controllers. To begin controlling the helicopter using the ROSH and RTSSH VR controllers, touch them with your hand, press the grip button on the controller, and then release it. Your hand will remain engaged and control the helicopter's movement. To detach from the ROSH or RTSSH, press the grip button again.

And in order to control the pedals without a joystick, you need to turn on the helicopter pedal control main menu (on the right hand) -> menu item Pedal control off. To turn it off, you need to click there again.

To enable control of the helicopter levers - the joystick, go to the main menu (on the right hand) -> menu item - Joystick is off.
To turn it off, you need to click there again.